Measure Theoretic Probability
Lesson 1
Sample Spaces, Events, Fields and σ-Fields

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Prerequisites

What this course presumes

You have already taken a regular probability course, where you dealt with:

- Set theory
- Random variables
- Distributions such as the **Binomial** and the **Exponential**;
- Associated functions known as **PDFs** (Probability Density Functions).

Measure Theory Is a Language

The language analogy

Think of a culture with a language different from yours. It has an expression for something that **just does not translate**, or the other way round. Some things are easier to say in one language than another.

Measure theory = a language for probability

- Some things become **easier to say** with it.
- Some things are **impossible to say** without it.

Selling point: Discrete and Continuous Are Unified

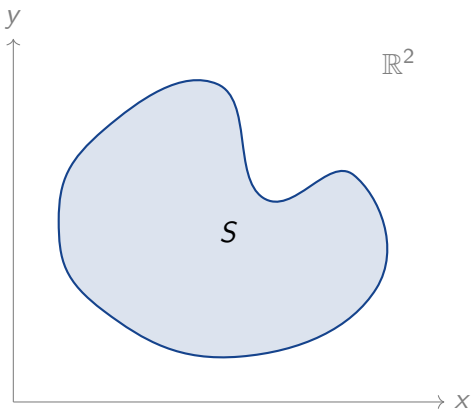
In a regular probability course

Discrete	sums $\sum \ldots$"
Continuous	integrals $\int$

With measure theory

There is **no need to split into cases**. The treatment of discrete and continuous random variables and probabilities is unified.

Measure Theory Revolves Around Sets



Exercise

Here is an irregular region S sketched in $\mathbb{R}^2$, the plane.

How do you Measure this set.

Some Scenarios

1. You try to find the **area**, which is hard because the set is not regular.
2. You ask: **what do you mean by "measure"?**

What Does "Measure" Mean?

Familiar ways to measure

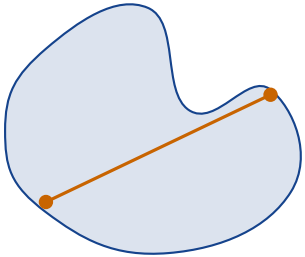
Dimension	Measure
1D	length
2D	area
3D	volume
nD	n-dimensional version of volume

These are known as the Lebesgue measure of the set.

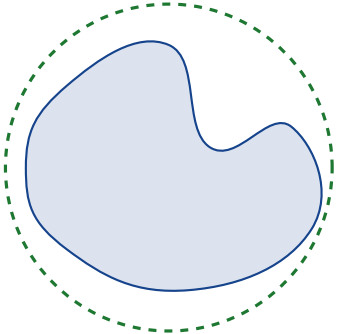
Key problem

Area is not the only option. **We need to define what "measure" means.**, we will look at that soon.

Other Ideas for "Measuring" a Set



Idea 1: length of the longest line joining two points of the set and *completely contained* in it.



Idea 2: area of the smallest circle in which the set can be inscribed.

Before we can say which of these "count" as measuring

we must first define what is meant by measure.

The Space Ω

Setting

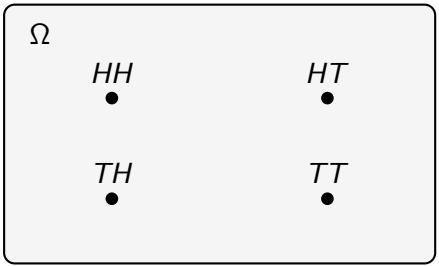
Let Ω be a **non-empty set**.

Lets Link back to a regular probability case

Ω is like the **sample space** of an experiment: the set of all possible outcomes.

Flip a coin twice (fair or not)

$\Omega = \{HH, HT, TH, TT\}$, so there are **4 outcomes**.



In this course Ω is abstract and usually not tied to an experiment (there need not be a coin). Relating each idea to regular probability will help **ground us**.

Events Are Subsets of Ω

Setting

Let A be a **subset** of Ω, written $A \subseteq \Omega$.

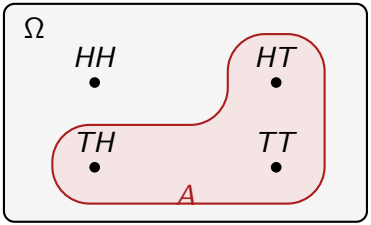
In a regular probability

A represents an **event**.

Event: "at least one tail"

$$A = \{HT,\ TH,\ TT\}$$

Three outcomes out of four in total.



Abstract setting

Ω is just a non-empty set and A is just a subset of it. There isn't a coin, but we can still relate things back to one.

Probability of an Event

Fair coin: equally likely outcomes

$A = \{HT, TH, TT\}$ is the event "at least one tail":

$$\mathbb{P}(A) = \frac{3}{4}.$$

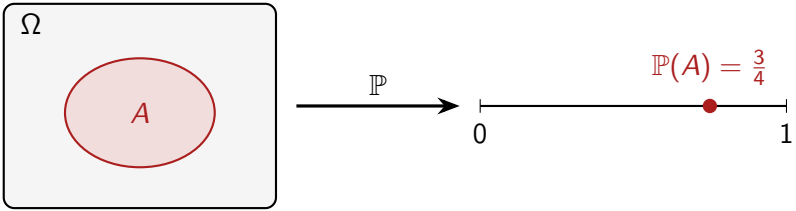
What does "A happens" mean?

Perform the experiment **one time**. A happens exactly when your outcome is in the set A. The probability that the outcome lies in A (3 of 4 equally likely things) is $3/4$.

In the abstract setting

$\mathbb{P}$ is **its own thing**, outside of any experiment: a function that takes in subsets of Ω and returns numbers between 0 and 1.

$\mathbb{P}$ Is a Function on Sets



This implies

$$\mathbb{P}: \ \{\text{subsets of } \Omega\} \longrightarrow [0,1].$$

The Power Set in the Coin Example

The power set

The **power set** of Ω is the set of all possible subsets of Ω.

Type of subset	Count	Example
Empty set	1	$\emptyset$
Singletons	4	$\{HH\}$
Pairs	6	$\{HH, HT\}$
Triples	4	$\{HT, TH, TT\}$
Whole set	1	Ω

Total: $16 = 2^4$ subsets.

The empty set

$\emptyset$ is the event that **nothing happens**.
When you flip the coin twice, something *has* to happen, so

$$\mathbb{P}(\emptyset) = 0.$$

The Problem and the Goal

Problem

We could go through every set in the coin experiment and assign probabilities, but this is **tedious** for bigger sets. For an **uncountably infinite** set it is awful, in fact impossible.

Goal

Assign $\mathbb{P}$ to **some** subsets of Ω, which can then be used to extend $\mathbb{P}$ to other subsets of Ω in a kind of natural way.



The Subsets We will Look At

Setting

- Let Ω be a **non-empty set**. (This is the line that precedes everything we do.)
- Let $\mathcal{F}$ be a **collection of subsets** of Ω.

Remark

$\mathcal{F}$ **does not have to be all subsets**. It does not have to be the power set.
We will want $\mathcal{F}$ to be something called a field (equivalently an algebra).

Field (Algebra)

Definition

A collection $\mathcal{F}$ of subsets of Ω is a **field** (or **algebra**) if:

(i) $\Omega \in \mathcal{F}$;

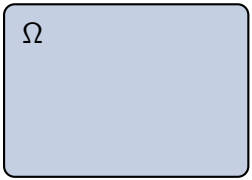
(ii) **closed under complements**: if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;

(iii) **closed under finite unions**: if $A_1, \ldots, A_n \in \mathcal{F}$, then $A_1 \cup \cdots \cup A_n \in \mathcal{F}$.

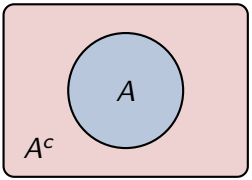
Remark on (i)

Members of $\mathcal{F}$ are **subsets** of Ω. Since every set is a subset of itself, $\Omega \subseteq \Omega$, so Ω is allowed to be a member of $\mathcal{F}$. Here "subset" means $\subseteq$ (it may be **all** of Ω), not a *proper* subset $\subsetneq$.

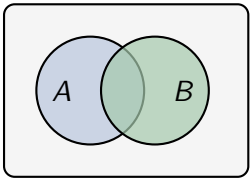
The Three Properties Pictured



(i) $\Omega \in \mathcal{F}$



(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$



(iii) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$

Checking (iii): Two Sets Are Enough

Claim

To show $\mathcal{F}$ is closed under **finite** unions, it is enough to show that for all $A, B \in \mathcal{F}$,

$$A \cup B \in \mathcal{F}.$$

Take two sets at a time

For three sets $A, B, C \in \mathcal{F}$: since $A \cup B \in \mathcal{F}$, we are left with two sets $A \cup B$ and C, so

$$A \cup B \cup C = (A \cup B) \cup C \in \mathcal{F}.$$

In general,

$$A_1 \cup \cdots \cup A_n = \left(A_1 \cup \cdots \cup A_{n-1}\right) \cup A_n,$$

and we reach a union of n sets by taking two at a time.

Equivalent Definition 1: Use $\emptyset$ Instead of Ω

Replace (i) by (i′): $\emptyset \in \mathcal{F}$

The resulting definition describes exactly the same collections.

Why they are equivalent (using closure under complements)

- If $\Omega \in \mathcal{F}$, then $\emptyset = \Omega^c \in \mathcal{F}$.
- If $\emptyset \in \mathcal{F}$, then $\Omega = \emptyset^c \in \mathcal{F}$.

The complement of $\emptyset$ with respect to Ω is Ω, and vice versa.

Conclusion

It is equivalent to have **either** $\emptyset$ **or** Ω as property number one.

Equivalent Definition 2: Intersections Instead of Unions

Replace (iii) by (iii′): closed under finite intersections

Again this describes exactly the same collections, by De Morgan's laws.

Unions $\Rightarrow$ intersections

Take $A, B \in \mathcal{F}$. Then $A^c, B^c \in \mathcal{F}$ (closed under complements), so $A^c \cup B^c \in \mathcal{F}$. But

$$(A \cap B)^c = A^c \cup B^c \quad \Longrightarrow \quad A \cap B = \left(A^c \cup B^c\right)^c \in \mathcal{F}.$$

Intersections $\Rightarrow$ unions

Similarly, $A \cup B = \left(A^c \cap B^c\right)^c \in \mathcal{F}$.

σ-Field

Definition: σ-field (σ-algebra)

A collection $\mathcal{F}$ of subsets of Ω is a **σ-field** if:

(i) $\Omega \in \mathcal{F}$;

(ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;

(iii$^\sigma$) **closed under countable unions**: if $A_1, A_2, A_3, \cdots \in \mathcal{F}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

The "σ" pattern

This is not an equivalent definition but a **new** one. "Countable" means possibly infinite, and a finite union is a countable union. In this course we define something using a *finite* action, then **elevate** it to a *countable* number of things, and put a σ in front.

Example 1: The Trivial σ-Field

Let Ω be any non-empty set and

$$\mathcal{F} = \{\emptyset,\ \Omega\}.$$

Why it is a σ-field

(i) It contains Ω.

(ii) **Complements:** $\Omega^c = \emptyset$ and $\emptyset^c = \Omega$.

(iii$^\sigma$) **Countable unions:** if every set is $\emptyset$, the union is $\emptyset$. If *at least one* set is Ω, the union is Ω. So any countable union stays in $\mathcal{F}$.

This is known as the trivial σ-field.

Example 2: The Power Set

Let Ω be a set and

$$\mathcal{F} = \mathcal{P}(\Omega) = \{\text{all subsets of } \Omega\}.$$

Why it is a σ-field

- Any subset of Ω has a complement that is **necessarily another subset of** Ω.
- Any countable union of subsets of Ω is again a subset of Ω.
- Ω itself is a subset of Ω.

Everything is "already in there by definition".

Example 3: The σ-Field Generated by A

Take any subset $A \subseteq \Omega$ and put

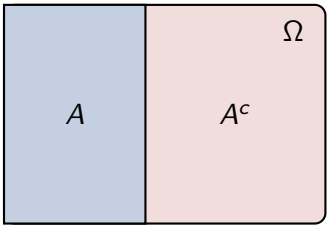
$$\mathcal{F} = \{ A,\ A^c,\ \emptyset,\ \Omega \}.$$

It is a (small) σ-field

(i) It contains Ω.

(ii) Complements: take any of the 4 sets, its complement is in the list.

(iii$^\sigma$) Any countable union of sets in the list is again in the list.



Important

This is the σ-field generated by A, and it will be an important σ-field for us throughout the course.

Example 3: Checking Closure Under Unions

Complements

set	complement
$\emptyset$	Ω
Ω	$\emptyset$
A	A^c
A^c	A

Unions

Whatever countable collection of these four sets you take, the union is one of the four:

- contains Ω, or contains both A and A^c $\Rightarrow$ union $= \Omega$;
- otherwise only A (and maybe $\emptyset$) $\Rightarrow$ union $= A$;
- otherwise only A^c (and maybe $\emptyset$) $\Rightarrow$ union $= A^c$;
- only $\emptyset$ $\Rightarrow$ union $= \emptyset$.

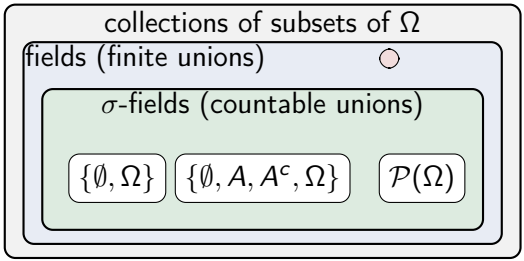
Every σ-Field Is a Field

Fact

Closed under countable unions includes closed under **finite** unions (countable can be countably infinite, *but it can also be finite*). So every σ-field is automatically a field.

Not the other way around

A field need not be a σ-field. We now build an example.



A Field That Is Not a σ-Field: Definition

Let $\Omega = \mathbb{R}$, the real line, and define $\mathcal{F}$ as follows

$\mathcal{F}$ consists of $\emptyset$ together with all **finite disjoint unions** of intervals of either form

$$(a, b] \qquad \text{or} \qquad (a, \infty)$$

where $a < b$ and a may be $-\infty$.

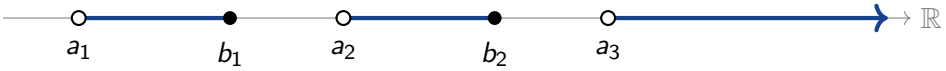
Check (i)

$(-\infty, \infty) = \mathbb{R}$ is of the form (a, ∞) with $a = -\infty$. So $\mathbb{R} \in \mathcal{F}$ and property (i) holds.

Plan

Show (ii) closure under complements, discuss (iii) closure under finite unions, then find a countable union that leaves $\mathcal{F}$.

What the Sets in $\mathcal{F}$ Look Like



A typical set in $\mathcal{F}$

$$(a_1, b_1] \cup (a_2, b_2] \cup (a_3, \infty)$$

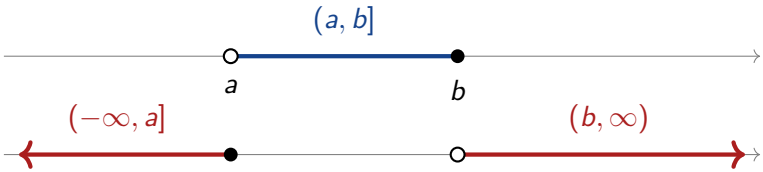
Closed Under Complements (a)

Take the simplest set in $\mathcal{F}$: $(a, b]$

Its complement is a union of **two intervals**:

$$(a, b]^c = (-\infty, a] \cup (b, \infty),$$

and both pieces are of the allowed form, so the complement is in $\mathcal{F}$.

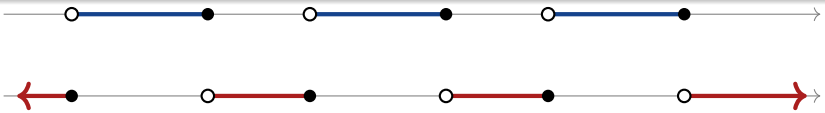


Closed Under Complements (b): Finite Disjoint Unions

Case 1: a finite union of half-open, half-closed intervals

$$A = \bigcup_{i=1}^{n} (a_i, b_i] \implies A^c = (-\infty, a_1] \cup (b_1, a_2] \cup \cdots \cup (b_n, \infty)$$

This is again a finite union of allowed intervals, so $A^c \in \mathcal{F}$. (If $a_1 = -\infty$ the first piece is empty and simply disappears.)



Exercise

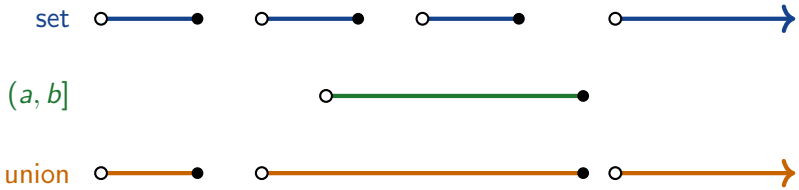
Check the **other cases**: sets of the form (a, ∞), and sets that are a union of both kinds.

Closed Under Finite Unions: An Example

Idea

It is enough to consider unions of two sets, but here the sets can be of **different kinds**, so several cases arise. Here is one complicated set in $\mathcal{F}$ and its union with another set of $\mathcal{F}$:

$$\underbrace{(a_1, b_1] \cup \cdots \cup (a_n, b_n] \cup (a_{n+1}, \infty)}_{\text{set in } \mathcal{F}} \quad \cup \quad \underbrace{(a, b]}_{\text{set in } \mathcal{F}}$$



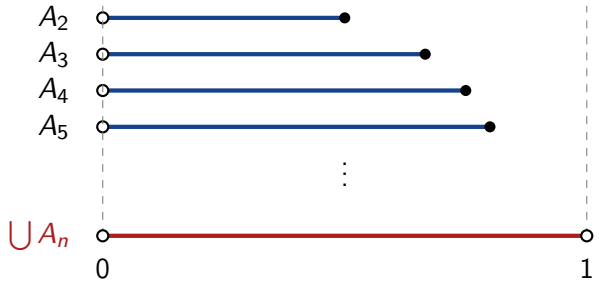
The union is still a finite union of allowed intervals, so it is in $\mathcal{F}$.

Why $\mathcal{F}$ Is Not a σ-Field

A countable collection in $\mathcal{F}$ whose union leaves $\mathcal{F}$

$$A_n = \left(0,\ 1 - \tfrac{1}{n}\right], \qquad n = 1, 2, 3, \ldots$$

Each $A_n \in \mathcal{F}$: $A_1 = (0, 0] = \emptyset$, $A_2 = (0, \frac{1}{2}]$, $A_3 = (0, \frac{2}{3}]$, $A_4 = (0, \frac{3}{4}]$, $\ldots$



Why $\mathcal{F}$ Is Not a σ-Field (continued)

The union

Each interval is contained in the next, so the union keeps stretching and **never quite reaches** 1:

$$\bigcup_{n=1}^{\infty} A_n = (0,1) \qquad \text{(open interval from 0 to 1).}$$

Conclusion

$(0,1)$ is **not one of the sets in our definition of** $\mathcal{F}$. (A bounded set in $\mathcal{F}$ must be closed on the right at its upper endpoints, but 1 is excluded here.) Hence

$\mathcal{F}$ is a field but not a σ-field.

What If We Throw That Set Back In?

Building up a field or σ-field

Define $\mathcal{F}$ as before, find sets that **violate** being a field or σ-field, create the new sets that should be in there but are not, and **throw them back in**. We can keep building up a field or a σ-field this way.

Dictionary: Regular Probability $\leftrightarrow$ Measure Theory

Grounding the abstract objects

Regular probability	Measure theory	Example (two coin flips)
sample space	non-empty set Ω	$\{HH, HT, TH, TT\}$
event	subset $A \subseteq \Omega$	$A = \{HT, TH, TT\}$
"A happens"	outcome lies in A	$\omega \in A$
probability of event	$\mathbb{P}$: subsets $\to [0,1]$	$\mathbb{P}(A) = 3/4$
impossible event	$\emptyset$	$\mathbb{P}(\emptyset) = 0$

Lesson 1: Summary

Definitions

- **Field (algebra):** $\Omega \in \mathcal{F}$ (or $\emptyset \in \mathcal{F}$), closed under complements, closed under **finite** unions (or finite intersections).
- **σ-field (σ-algebra):** same, but closed under **countable** unions.

Examples

Trivial $\{\emptyset, \Omega\}$; power set $\mathcal{P}(\Omega)$; generated by A: $\{\emptyset, A, A^c, \Omega\}$.

Key facts

Every σ-field is a field, but **not conversely**: finite disjoint unions of $(a, b]$ and (a, ∞) on $\mathbb{R}$ form a field, yet $\bigcup_n (0, 1 - \frac{1}{n}] = (0, 1) \notin \mathcal{F}$.